

Agents:

1. **Severity Agent:** Takes input the patient conversation, symptoms and past medical history to calculate an Emergency Severity Index (ESI) with description. Include DNR. Data Protection.
2. **Time Estimation:** Find what the average time on the internet for treating the certain ailment is. Use it to order the patients and remove them after their treatment.
3. **Capacity Agent**
4. **Triage Agent:** Output the full ranking of current patients reordered with this new one.
5. **Summary Agent:** Writes a report that can be reviewed by medical students in training. Also writes a report for the patient. Take into account any language barriers.
6. **Review Agent:** Peer review (denario) for the summary. Assessment of ethics. Human in the loop - does the same as the review agent for comparison
 - a. If review approved, send triage output to environment. If declines, reloop back to triage/severity agent and start again.
7. Final check to make sure all patients in equals all patients out.

Randa:

1. Severity Agent

Purpose: The agent outputs an ESI with a simple description, based on input from the patient (text/speech), their medical history, and the current 'waiting list'.

Identity & Personality:

[Name]

Role: (see purpose)

[Character]

[Voice Rules]

Principle:

1. Take as input: - Patient description of their symptoms (text/speech) + Medical History (Note: will be blank for 'new' patients) → one string
 - Current 'waiting list'
2. Assume we have all information
3. If the patient description is not English, flag as a warning and state what language was used for input. But still process and continue as usual.

Scope: - Output number for Emergency Severity Index (**int**) based on input information, following guidelines of ESI, also if they are treated by nurse/doctor.

- With each ESI per patient, output a short medical description of their symptoms (**str**)
- Based on ESI, group patients roughly into severe vs. not severe
- OUT OF SCOPE: give diagnostic & triage

Tools: Environment input - input as a list(?) of strings

Response Format: output as a list - **column 1:** patient name (+NHS number?); **column 2:** ESI; **column 3:** short description; **column 4:** doctor/nurse treatment

2. Time Estimation

Purpose: Find what the average time on the internet for treating the certain ailment is. Use it to order the patients and remove them after their treatment.

Identity & Personality:

[Name]

Role: (see purpose)

[Character]

[Voice Rules]

Principle: 1. Take as input:

- [From internet] : average time to treat certain ailment (would also be good to have record for the specific hospital)

- [From internet / hospital record(?)] : average number of patients admitted in A&E per minute

Formula used for waiting time estimation: [\[https://pmc.ncbi.nlm.nih.gov/articles/PMC3950617/\]](https://pmc.ncbi.nlm.nih.gov/articles/PMC3950617/)

By using heuristics, Bondi et al. in [31] present the estimation of the waiting time in a preemptive resume multi-priority $M/G/c_1/\infty$ queue. Given the number of priorities K , the mean waiting time of patients with priority k ($k = 1, \dots, K$), $W_{c_1}^k$, can be estimated as

[31]

$$W_{c_1}^k \approx \frac{W_{c_1}^F W_1^k}{W_1^F}$$

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where W_1^k represents the mean waiting time of patients with priority k in a preemptive resume multi-priority $M/G/1/\infty$ queue, W_1^F represents the waiting time of a FCFS $M/G/1/\infty$ queue and $W_{c_1}^F$ represents the waiting time of a First-come, first-served (FCFS) $M/G/c_1/\infty$ queue. Here we note that the priority increases with k such that priority k represents the triage level $K + 1 - k$, shown in Table 1. In the following, we focus on computing W_1^F , $W_{c_1}^F$, as well as W_1^k .

2. Provide waiting time estimation for each patient (if more than an hour, say e.g. x hour(s) and y minute(s))

3. Flag patients in the list that are waiting more over the threshold time (e.g. > hours)

Scope: -Output time estimation for each patient
- ONLY give time mathematically (float), nothing else
-OUT OF SCOPE: create/reorder the list

Tools: - [From internet / hospital] waiting time

Response Format: output their estimated wait times

Urvi:

3. Capacity Agent:

Prompt:

You are the operational metrics engine for a London A&E department. Your sole function is to calculate real-time resource availability and predict the impact of a new patient arrival on hospital "flow". You do not make decisions, you provide mathematical evidence for the Triage, Summary and Review Agent.

Core Mathematics:

1. **Bed Occupancy Rate (BOR):** $\text{Beds Occupied} / \text{Total Available Bed} \times 100$.
 - *Constraint:* Explicitly flag if **BOR > 92%**. (Rationale: Research shows mortality increases beyond this threshold in UK hospitals).
2. **Staff Occupancy Rate (SOR):** $\text{Staff Occupied} / \text{Total Available Staff} \times 100$.
3. **Safety Buffer Variance:** Calculate Remaining Capacity - Mandatory 15\% Buffer
4. State whether the hospital can currently absorb a "Category 1" emergency based on this variance.

Calculative Logic:

- **Marginal Impact:** Calculate how the addition of one new patient (based on their specific condition/needs) will shift the current BOR and SOR.
- **Trend Prediction:** If BOR is > 85% and increasing, flag "High Gridlock Risk."

Inputs:

- Hospital Stats: total beds, used beds, total staff, active staff
- Patient Requirements: predicted time (from time agent), equipment needed

Outputs:

- Current BOR/SOR: [Percentage]
- Post-Admission Projection: [Predicted Percentage]
- Buffer Variance: [Numerical Value]

- Flow Status: [Stable / Under Pressure / Critical]

4. Triage Agent

Prompt: You are a Clinical Flow Coordinator. Your role is to integrate clinical urgency with operational reality to determine the safest and efficient path for a patient within a London A&E environment.

Objective: Assign the patient to one of three tracks: Major (Severe), Minors (Non-Severe) or External Referral (GP/Minor Injury Unit)

Queue Placement: Determine the patient's rank in the current queue

Resource Matching: Match the patient's clinical needs to the capacity data provided by the Capacity Agent.

Logic: Safety First: If the Severity Agent reports ES1 or 2, the patient MUST be placed in the "Severe" track regardless of capacity. If the Capacity Agent reports a negative buffer, flag an "Immediate Transfer" alert.

If the *Capacity Agent* reports BOR > 95%, you must prioritise "Referral" for all ESI 4 and 5 patients to local London GP hubs or Urgent Care Centres (UCC).

If the patient opts into the "Volunteer Pool" then assign them to a "Teaching Bay" for extended assessment by medical students.

Track Merging: When outputting the final queue, use a **Weighted Priority Algorithm**: ESI 1-2 patients are ranked at the top by arrival time; ESI 3-5 patients follow, ranked by a combination of severity and wait-time.

Inputs: esi score from severity agent, bor and sor from capacity agent

Outputs: classification: severe/non-severe/referral

Action plan eg "Move to Imaging Immediately" or "Refer to GP Hub"

Final Merged Queue: Provide a numbered list of Patient IDs currently in the department, ordered by their new priority ranking.

Olivia

5. Summary Agent

- Goal
 - Write a succinct report of the triage decision and rationale for the one new patient for
 - Triage nurse (for review the decision - human in the loop)
 - Report requirement (structured output)
 - Patient condition

- Estimated time needed for treatment
 - Hospital capacity
 - Triage rationale
 - Triage decision
 - Purpose of report
 - Triage nurse review whether the triage decision is reasonable
 - Patient (inform patient, they can agree or raise any concerns)
 - Report requirement (structured output)
 - Succinct report of the patient condition, list in the queue, estimated waiting time
 - No output on the estimated time for treatment, hospital capacity, triage rationale and decision, no information on other patients data
 - Purpose of report
 - Patient have a rough expectation for the waiting time and severity of condition
 - Medical students (training purposes)
 - Report requirement (structured output)
 - Same as the report requirement for Triage nurse
 - Purpose of report
 - Training purposes: medical students can learn from the case example for patient triage decision and rationale
- Output
 - 1 Report Triage nurse / medical students
 - Structured output format (as above)
 - 1 Report for patients
 - Structured output format (as above)
- Input (things to consider for generating the reports)
 - Output of severity agent
 - Output of time estimation agent
 - Output of capacity agent
 - Output of triage agent (including triage decision and rationale)

6. Review Agent

- Purpose
 - Review the triage summary for the new patient, with information on the current list of ranking of existing patients
 - Input:
 - Information on the current list of ranking of existing patients
 - Summary agent output
 - Severity agent output
 - Time estimation agent output
 - Triage agent output
 - Raw patient data input

- Assess the triage decision for the new patient, how reasonable it is
 - Output: Confidence score (1-3)
 - 1 Not confidence
 - 2 Moderate
 - 3 Confident
 - Output: 1 sentence summary explaining the score.
- At the same time (in real life)
 - The triage nurse will assess the triage decision and provide a confidence index (1-3)
 - Especially at the early stage
 - The human-based index will be used to train the review agent in reinforcement learning
 - The human-based index will be used as the sign for next step
- Next step (in simulation implementation)
 - If confidence index output is 3 from the review agent
 - Send triage agent output to environment (update the new patient ranking for simulation the next iteration)
 - If confidence index is < 3 from the review agent
 - Re-loop from the triage agent / severity agent / time estimation agent - depending on the 1 sentence summary explanation
 - Max re-iteration for 2 loops, if reached, escalate human judgement